

Align on English anchors only; evaluate on held-out English only

A Pretraining and same-language alignment

Fit Procrustes maps using
3000 general-vocabulary English anchors

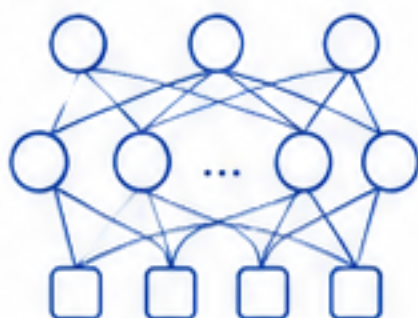
Bilingual EN+ZH

EN tokens + ZH tokens
(pretraining mixture)



Monolingual EN

EN tokens
(pretraining mixture)

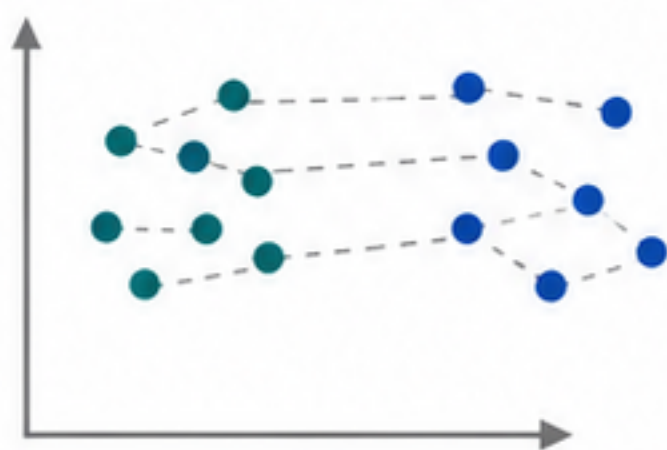


Bilingual EN+FR

EN tokens + FR tokens
(pretraining mixture)

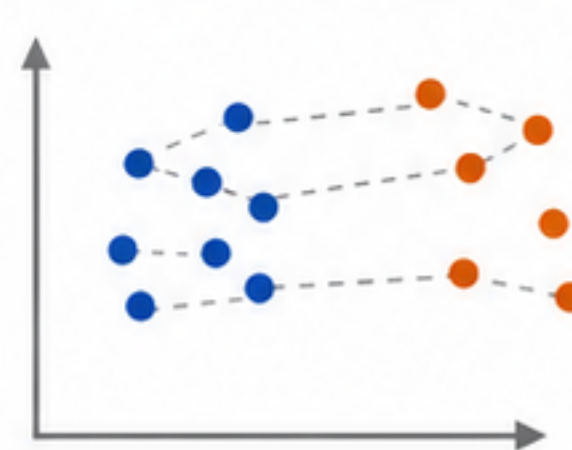


Aligned English subspace
(EN ↔ EN+ZH)



English-to-English
alignment only

Aligned English subspace
(EN ↔ EN+FR)



- Bilingual EN+ZH
- Monolingual EN (reference)
- Bilingual EN+FR
- Nearest-neighbor match after alignment

B Held-out English evaluation

Probe: wedding dress
1000 held-out English probes

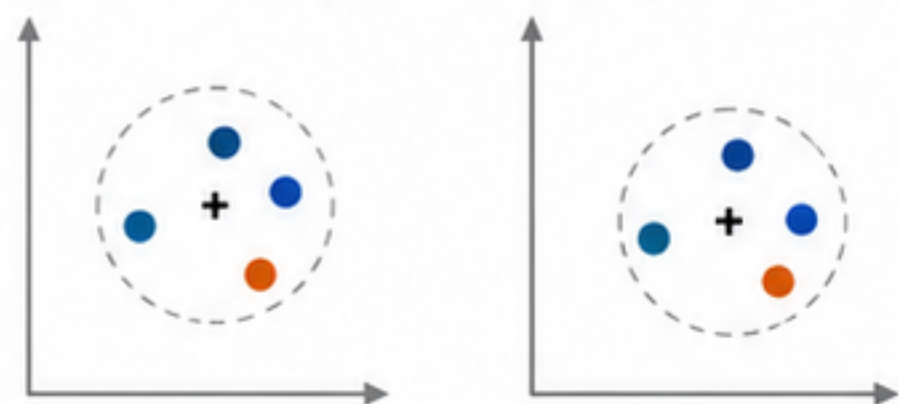
Axis: red ↔ white
50 semantic axes

1 Token embeddings

Aligned representations remain close.

Probe: "wedding dress" Axis: red ↔ white

- EN+ZH
- EN (ref)
- EN+FR



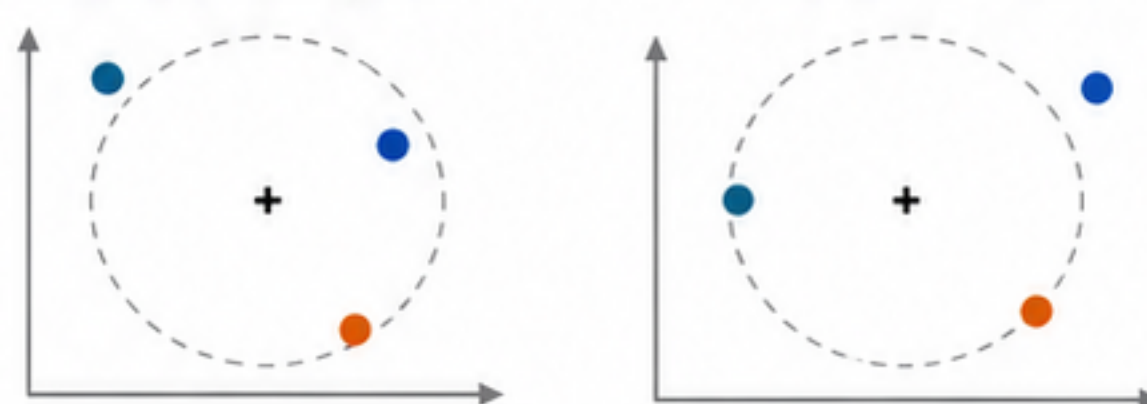
Small mismatch after alignment

2 Contextual states

Aligned representations diverge more.

Probe: "wedding dress" Axis: red ↔ white

Axis: red ↔ white



Larger mismatch after alignment

----- Dashed circle denotes average distance from EN reference (smaller is better)